



3.3 Non-right triangles: sine and cosine rules

Geometry and trigonometry · IB Diploma · Concept guide and original exam-style practice

Understand the concept

For any triangle with sides a, b, c opposite angles A, B, C : the sine rule $a/\sin A = b/\sin B = c/\sin C$ relates sides and opposite angles (used when given two angles and a side, or two sides and a non-included angle — watch for the ambiguous case). The cosine rule $a^2 = b^2 + c^2 - 2bc \cdot \cos A$ (and rearrangements) is used when given three sides, or two sides and the included angle. Area = $(1/2)ab \cdot \sin C$ using two sides and the included angle.

Worked example: follow the reasoning

In triangle ABC, $a=7$, $b=9$, $C=40^\circ$. Find c and the area.

Cosine rule: $c^2 = 7^2 + 9^2 - 2(7)(9)\cos(40^\circ) = 49 + 81 - 126(0.766) = 130 - 96.5 = 33.5$

$c = \sqrt{33.5} = 5.79$ (3 s.f.)

Area = $(1/2)(7)(9)\sin(40^\circ) = 31.5(0.643) = 20.3$ (3 s.f.)

Quick check · multiple choice

In a triangle, $a=5$, $b=8$, $A=30^\circ$. Using the sine rule to find angle B , how many possible values of B ($0^\circ < B < 180^\circ$) are there?

- A 0
- B 1
- C 2
- D 3

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 In triangle PQR, $PQ=12$ cm, $QR=15$ cm, and angle $PQR=110^\circ$. [8]

- (a) Find the length PR, correct to 3 significant figures.
- (b) Find angle QPR, correct to the nearest degree.
- (c) Find the area of triangle PQR.

Final answer: _____



Answers and explanation

Question 1 [8]

(a) Cosine rule: $PR^2 = 12^2 + 15^2 - 2(12)(15)\cos(110^\circ) = 144 + 225 - 360(-0.342) = 369 + 123.1 = 492.1$.
 $PR = \sqrt{492.1} = 22.2$ cm.

(b) Sine rule: $\sin(QPR)/15 = \sin(110^\circ)/22.2 \rightarrow \sin(QPR) = 15(0.9397)/22.2 = 0.6349 \rightarrow$
 $QPR = \arcsin(0.6349) = 39.4^\circ \approx 39^\circ$.

(c) Area = $(1/2)(12)(15)\sin(110^\circ) = 90(0.9397) = 84.6$ cm².

Quick check answer

C. $\sin B = (8\sin 30^\circ)/5 = 0.8$, giving $B = \arcsin(0.8) = 53.1^\circ$ or $B = 180 - 53.1 = 126.9^\circ$; both are valid since $a < b$ (ambiguous case).

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

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